

towards a ubi-comp future and also spearheading a lot of practical implementations. Ubi-comp has also been included in the developmental roadmaps of these countries. Most countries are fiddling around with smart electricity grids for improved power management and conservation—one of the smartest examples of ubiquitous computing. India too has a wing of its Centre for Development of Advanced Computing (C-DAC) doing research on ubi-comp.

"In Japan, the Ministry of Internal Affairs and Communication has taken the lead in introducing ubi-comp technology. The Tokyo Metropolitan Government has shown strong interest in using the ubi-comp technology for sightseeing guidance, helping the aged and the physically-challenged to go around the town, and for maintaining public objects in parks and along highways, etc," says Ishikawa.

Another example is the Indian Ocean Tsunami warning system, which was installed after the tsunami of 2004. It includes a network of deep ocean sensors and 25 seismographic information stations relating information to 26 national tsunami information centres.

In China, there is a lot of government pressure on industries and local governments to implement various aspects of ubi-comp. "The city of WuXi is reportedly becoming the 'centre for sensing China' with various IoT pilot initiatives in progress. The country's first intelligent transformer substation for state grid went live in the city, fully automated with sensors, in early 2011. At the Shanghai World Expo 2010, 70 million tickets with RFID chips were integrated with three million cellphones and ubi-comp applications demonstrated for vehicles, rubbish management and food safety. Chongqing, having a

population of 32 million, is building the world's largest ubi-comp video surveillance system with at least half a million cameras to facilitate public safety management integrated into the city's municipal services, transportation, environmental monitoring and emergency response systems," says Arora.

At the industry level, there are several organisations like the Cisco co-founded Internet Protocol for Smart Object Communications (IPSO) Alliance, the ZigBee Alliance, IPv6 Forum, and standards bodies like the Institute of Electrical and Electronics Engineers (IEEE), the European Telecommunications Standards Institute (ETSI) and the International Telecommunications Union (ITU) that are contributing towards the development of ubi-comp.

Ease my fears, please

All said, for any technological revolution to be successful, the people at large need to be comfortable with it. As far as ubi-comp goes, there is a serious fear of privacy loss that continues to linger in the minds of people. As of now, we are nowhere close to settling these fears. Nations such as the European Union members are struggling to find a solution to these privacy concerns, tackling the problem head-on and inviting solutions from the public.

"What our team leader Prof. Ken Sakamura has been saying is that the society needs a new set of laws and regulations to deal with issues such as risks and privacy concerns brought about by the new technology. Take, for example, the privacy concerns. People in ordinary times don't want their whereabouts and IDs to be known to third parties. However, during emergencies, such as immediately after a typhoon or a big earthquake, people may want to transmit their whereabouts or

IDs to relief workers, especially medical staff, so that their loved ones can learn the status of each other. But where do you draw the line? Under what conditions is such capturing of whereabouts and IDs considered okay and when not?

"Cars kill people in traffic accidents. Insurance helps ease the situation to an extent. Telephones have been used by kidnappers. Wire-tapping helps find the kidnappers. So the society has found ways to cope with new risks brought about by the new technologies as long as the usage of the new technologies turned out to be a net profit for the society," explains Ishikawa.

"Apart from public concern over infringement of personal privacy, the governments may also have certain concerns over national security. Unless understood and addressed, potential personal privacy invasion issues could lead to overprotective policies or outright prohibition in countries and across international boundaries. Using sensors to protect critical infrastructure may raise security concerns over the possibility of unauthorised or hostile monitoring, or even a seizure of control over vital national assets. End-to-end security and management capability are therefore critical to the IoT," adds Arora.

Governments will also have to look into any related health concerns arising from, say, electromagnetic fields linked to sensors and wireless transmission or e-waste. Health and environmental concerns will gain greater prominence with increasing volume.

"None of these issues is insurmountable, but each must be taken into account from the start in the planning, design and strategy for any ubi-comp vision and deployment," concludes Arora. ●

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